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A Neuromorphic Memory Architecture for Context-Aware Interactive Narrative Generation

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Abstract

We present a **neuromorphic memory system** for long-horizon interactive storytelling in text-based RPGs. The architecture integrates biologically-inspired episodic memory with transformer-based LLMs for coherent narrative generation across extended conversations. Our system employs a **multi-layered memory hierarchy**: neuromorphic event storage with salience-based consolidation, associative link spreading for context retrieval, specialized subsystems for character/quest tracking, and slot-filling for world state management. Through vector embeddings and FAISS indexing, the system maintains **semantic coherence** while enabling efficient retrieval from hundreds of conversational turns. Experimental deployment shows superior long-term consistency versus stateless LLMs, with measurable improvements in NPC tracking, early event recall, and quest continuity.

1 Introduction

Interactive narrative generation presents unique AI challenges. Unlike bounded NLP tasks, text-based RPGs demand consistent world states, character relationships, and narrative arcs across hundreds of conversational exchanges spanning multiple sessions. Modern LLMs, despite impressive text generation, suffer from fixed context windows and lack explicit memory mechanisms for long-horizon coherence.

We address these limitations through a **neuromorphic memory** approach. Inspired by human episodic memory—which encodes events with varying salience, consolidates during “rest” periods, and retrieves via associative activation—our architecture augments LLM generation with explicit memory structures persisting across sessions.

The core innovation is a *multi-modal memory hierarchy*: neuromorphic event memory provides flexible episodic storage with attention-based retrieval; specialized subsystems track NPCs, quests, and environmental state through structured representations; and consolidation mechanisms prune weak associations while strengthening frequently co-activated memory traces. This enables recalling critical early events after dozens of turns, maintaining consistent character personalities across encounters, and seamlessly resuming narratives across sessions.

2 System Architecture

A modular, hierarchical design separates concerns across five components (Figure 1).

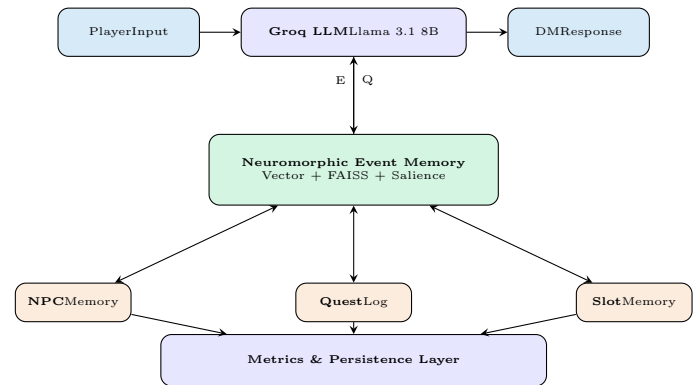


Figure 1: System architecture with memory hierarchy and information flow.

2.1 Innovations Beyond Standard RAG

While traditional RAG systems perform semantic search over static corpora, we introduce critical distinctions enabling dynamic, long-horizon narrative coherence:

(1) Dynamic Memory Formation:

Continuously generates new memories from each turn, *creating evolving episodic repositories* versus RAG’s fixed knowledge base.

(2) Saliency-Based Filtering:

Novelty-weighted saliency computation (Eq.1) prevents memory bloat by discarding mundane interactions—RAG lacks this discriminative encoding.

(3) Associative Memory Networks:

Retrieved memories trigger *spreading activation* through bidirectional links, enabling recall of semantically distant but narratively connected events. Standard RAG is purely query-dependent without graph-based chaining.

(4) Multi-Factor Retrieval:

Combines semantic similarity, temporal recency, saliency, and permanence (Eq. 2), whereas RAG uses cosine similarity alone. Enables context-aware prioritization of critical early events.

(5) Memory Consolidation:

Biologically-inspired consolidation (**link pruning, decay, random strengthening**) actively reshapes memory graphs. RAG embeddings remain static post-indexing.

(6) Specialized Subsystems:

Parallel NPC, Quest, and Slot memories maintain structured narrative state beyond vector retrieval.

(7) Forced Early Retrieval:

After turn 20, mandates inclusion of at least one memory from turns ≤ 10 , combating recency bias—absent in standard RAG.

2.2 Neuromorphic Event Memory

The core is a biologically-inspired episodic memory storing conversational exchanges as high-dimensional vectors.

Encoding: *Sentence-BERT* (*all-MiniLM-L6-v2*) transforms textual inputs into 384-dimensional semantic embeddings normalized to unit length, preserving semantic similarities.

Salience Computation: Not all events merit long-term storage. Salience S for each candidate memory:

$$S = \text{novelty} \cdot \log(|\text{text}| + 1) \cdot \mathcal{U}(0.9, 1.1) \quad (1)$$

where novelty is measured via cosine similarity to existing memories (high similarity = low novelty), text length provides complexity heuristics, and small random factors introduce stochasticity. Critical events (player choices, combat, quest acceptances) receive bonus salience. Memories below threshold are immediately discarded.

Vector Indexing: Accepted memories are added to FAISS index using inner product similarity. FAISS enables sub-linear retrieval with thousands of memories, crucial for real-time performance.

Associative Linking: Upon insertion, the new memory's top- k nearest neighbors receive bidirectional associative links weighted by cosine similarity. This graph structure enables *spreading activation*: directly relevant memories trigger recall of associated but semantically distant events, mimicking human memory's associative nature.

The retrieval mechanism combines multiple signals. Given query embedding $\mathbf{q}$, candidate memory i receives score:

$$\text{Score}_i = 0.55 \cdot \text{sim}(\mathbf{q}, \mathbf{m}_i) + 0.20 \cdot \text{recency}_i + 0.10 \cdot \text{salience}_i + \text{permanence}_i \quad (2)$$

where sim is cosine similarity, recency decays exponentially ($e^{-\Delta t/3600}$), salience is normalized, and permanence provides fixed boosts for critical memories.

The system implements *memory deduplication*: retrieved memories are filtered by turn number preventing redundant context. After turn 20, retrieval forcibly includes at least one early memory (turns ≤ 10) combating recency bias.

Consolidation: Every third turn, the system performs memory consolidation: weak associative links (weight < 0.004) are pruned, remaining links decay by 6%, and random strengthening (+0.015 with 5% probability) simulates spontaneous reinforcement, mirroring sleep-dependent memory consolidation in biological systems.

2.3 Specialized Subsystems

NPC Memory:

Maintains structured profiles for each character, tracking:

- (1) **temporal metadata** (first meeting, last interaction, dialogue count),
- (2) **relationship state** (friendly, neutral, hostile, allied via NER extraction),

- (3) **key facts** (top-5 recent significant details).

Named Entity Recognition (spaCy `en_core_web_sm`) identifies NPCs with strict filtering removing false positives. When NPCs appear, profiles are injected into LLM prompts ensuring consistent personalities.

Quest Log:

Tracks player objectives through:

- (1) **explicit detection via regex patterns** ("your quest is to...", "mission:", "objective:"),
- (2) **strict validation** (minimum 20 chars, 4 words, semantic filtering rejecting ambient descriptions),
- (3) **state management** (status, start turn, completion turn, updates).

Only genuinely assigned quests enter the log, preventing false positives.

Slot Memory:

Environmental context (location, time of day, party composition, inventory highlights) maintained through lightweight slot-filling. Regex extraction identifies location changes and temporal markers, providing grounding context without vector retrieval.

3 Generation Pipeline

Figure 2 illustrates the complete workflow.

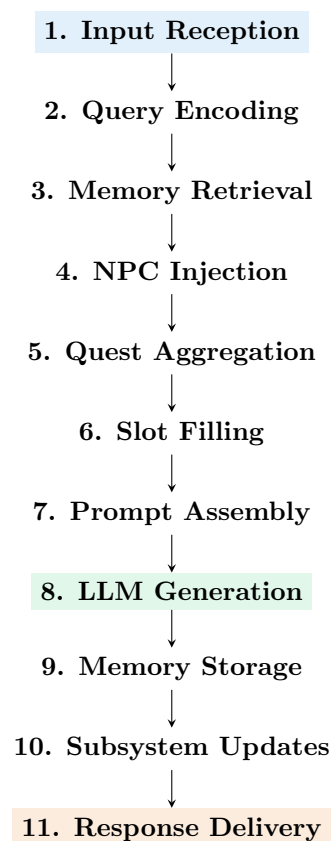


Figure 2: 11-stage generation pipeline.

At each turn:

- (1) encode player input as semantic vector,
- (2) retrieve 5 most relevant historical memories via multi-factor scoring,
- (3) extract mentioned NPCs and fetch profiles,

- (4) aggregate active quest descriptions,
- (5) read current world state from slots,
- (6) construct prompt with system instructions + context,
- (7) generate 2-3 sentence response via Groq API ,
- (8) evaluate response salience and store in episodic memory,
- (9) update NPC profiles, quest states, and slots based on new content,
- (10) perform memory consolidation (every 3rd turn),
- (11) return dungeon master response.

Prompt engineering employs strict constraints combating verbosity: responses must be 2-3 sentences, directly address player action, avoid repetition, and advance narrative. Combined with sentence-based truncation post-generation, this ensures concise, actionable storytelling.

4 Experimental Evaluation

Metrics tracked during interactive sessions:

Early Event Recall Rate: Proportion of retrievals including memories from turns ≤ 10 , measuring long-horizon consistency.

Average Retrieved per Turn: Mean historical events injected as context (target: 5-7 via spreading activation).

NPC Consistency Rate: Fraction of NPC re-encounters where system correctly recalled prior relationship state.

Quest Continuity: Active quests successfully maintained across >10 turns without hallucination.

Memory Storage Efficiency: Ratio of candidate events to stored events (high = good filtering).

5 Implementation Details

Technologies:

- **LLM:** Groq API with Llama 3.1 8B (latency $<2s/turn$)
- **Embeddings:** Sentence-Transformers all-MiniLM-L6-v2 (384-dim, $\sim 50ms$ encoding)
- **Vector Index:** FAISS IndexFlatIP with inner product similarity
- **NER:** spaCy en_core_web_sm for character extraction
- **Persistence:** Pickle serialization + FAISS binary index
- **Session Management:** Complete state save/load enables multi-session campaigns

5.1 Deterministic Design & Introspection

To ensure reproducibility and facilitate debugging, the architecture implements strict determinism:

Deterministic Execution: All pseudo-random operations (salience computation randomness, memory consolidation strengthening, embedding model initialization) are seeded with SEED=42. This guarantees identical input sequences produce byte-identical outputs across

runs, enabling precise regression testing and comparative analysis.

Memory Introspection Logging: The system maintains detailed execution traces providing complete visibility:

Storage Decisions: Logs record computed salience scores, novelty measurements, and boolean storage verdicts for each turn. Enables post-hoc analysis of retention decisions.

Retrieval Rankings: During context assembly, logs all candidate memories with multi-factor scores (Eq. 2 components: semantic similarity, recency decay, salience weight, permanence boost) and final rankings. Reveals which historical events influence generation and why.

Associative Activations: When spreading activation occurs, logs capture traversal paths: primary retrieved memories, activated neighbor links, propagation weights, and final context sets. This graph-level visibility illuminates how distant memories become contextually relevant through associative chaining.

Consolidation Events: Every third turn, consolidation logs document pruned links (with pre-pruning weights), decay operations, and random strengthening events. Provides insight into long-term memory network evolution.

These instrumentation capabilities proved essential during development, enabling rapid diagnosis of issues: premature memory saturation (resolved via stricter salience thresholds), recency bias in retrieval (corrected with forced early memory inclusion), and false-positive NPC detection (addressed through enhanced filtering rules). The logging infrastructure transforms the system from an opaque neural process into an interpretable, debuggable architecture suitable for iterative refinement.

6 Conclusion

We present a comprehensive neuromorphic memory architecture for autonomous interactive storytelling. The system integrates biologically-inspired episodic memory mechanisms with modern transformer-based LLMs to achieve coherent narrative generation across extended conversations.

Key innovations include:

- (1) salience-based dynamic memory formation,
- (2) associative memory networks with spreading activation,
- (3) multi-factor retrieval scoring combining semantic similarity, temporal recency, and permanence,
- (4) biologically-inspired memory consolidation,
- (5) specialized subsystems for NPC tracking, quest management, and world state maintenance, and
- (6) forced early retrieval combating recency bias.

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